



BITS Pilani presentation

BITS Pilani
Pilani Campus

Faculty Name
Faculty Department



EEE ZG512 / SS ZG512, EMBEDDED SYSTEM DESIGN

Lecture No. 05

ARM Features

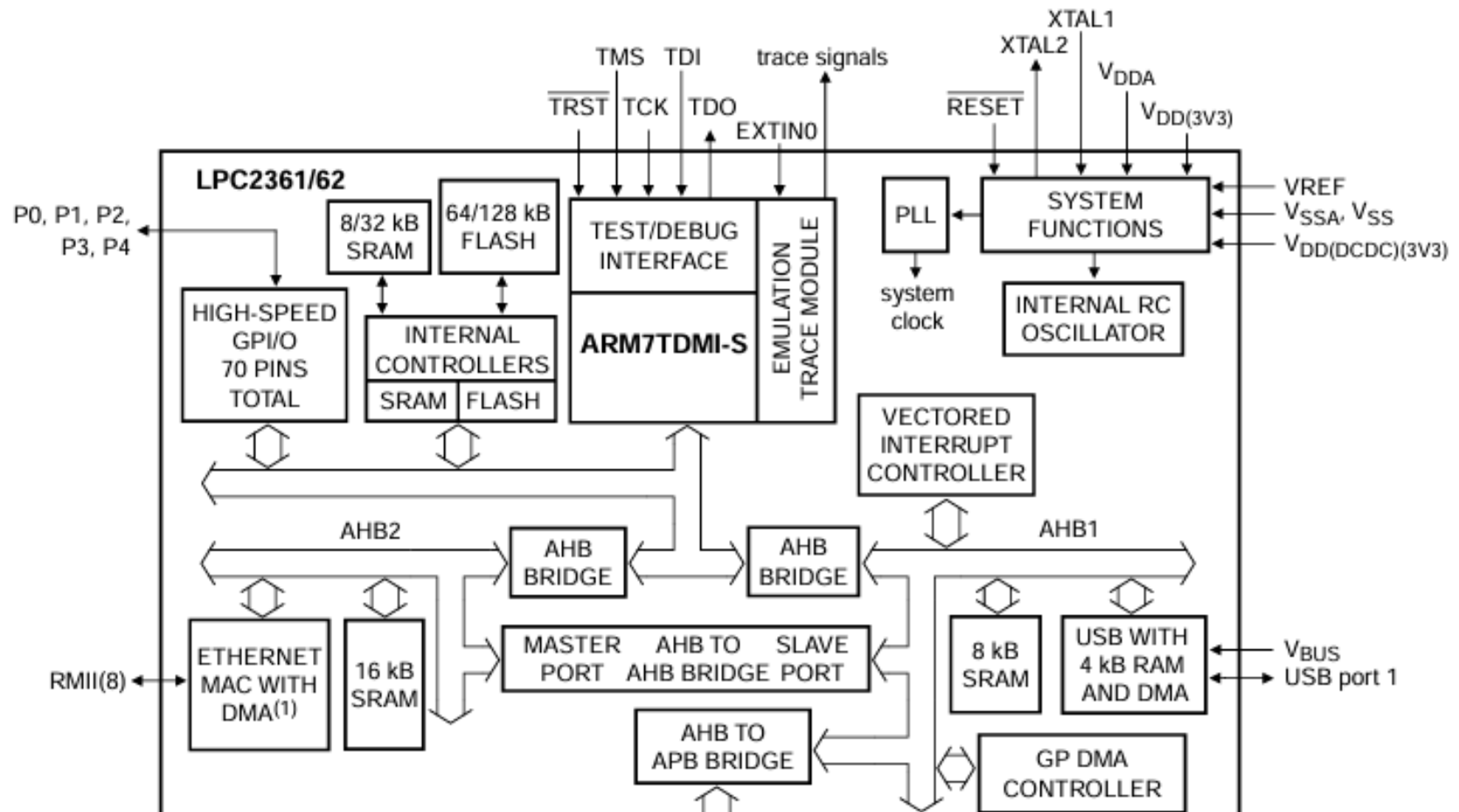
- Multiprocessing: Designed to process multiple processes simultaneously.
- Tightly Couple Memory: High access speed and less latency.
- MMU (Memory Management Unit) & MPU (Memory Protection Unit) : Improves efficiency and safety of memory units.
- Single cycle execution time: One instructions is executed each machine cycle.
- Thumb Technology: Extends 16 bit instructions to 32 bit instructions.

ARM Architecture

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lead

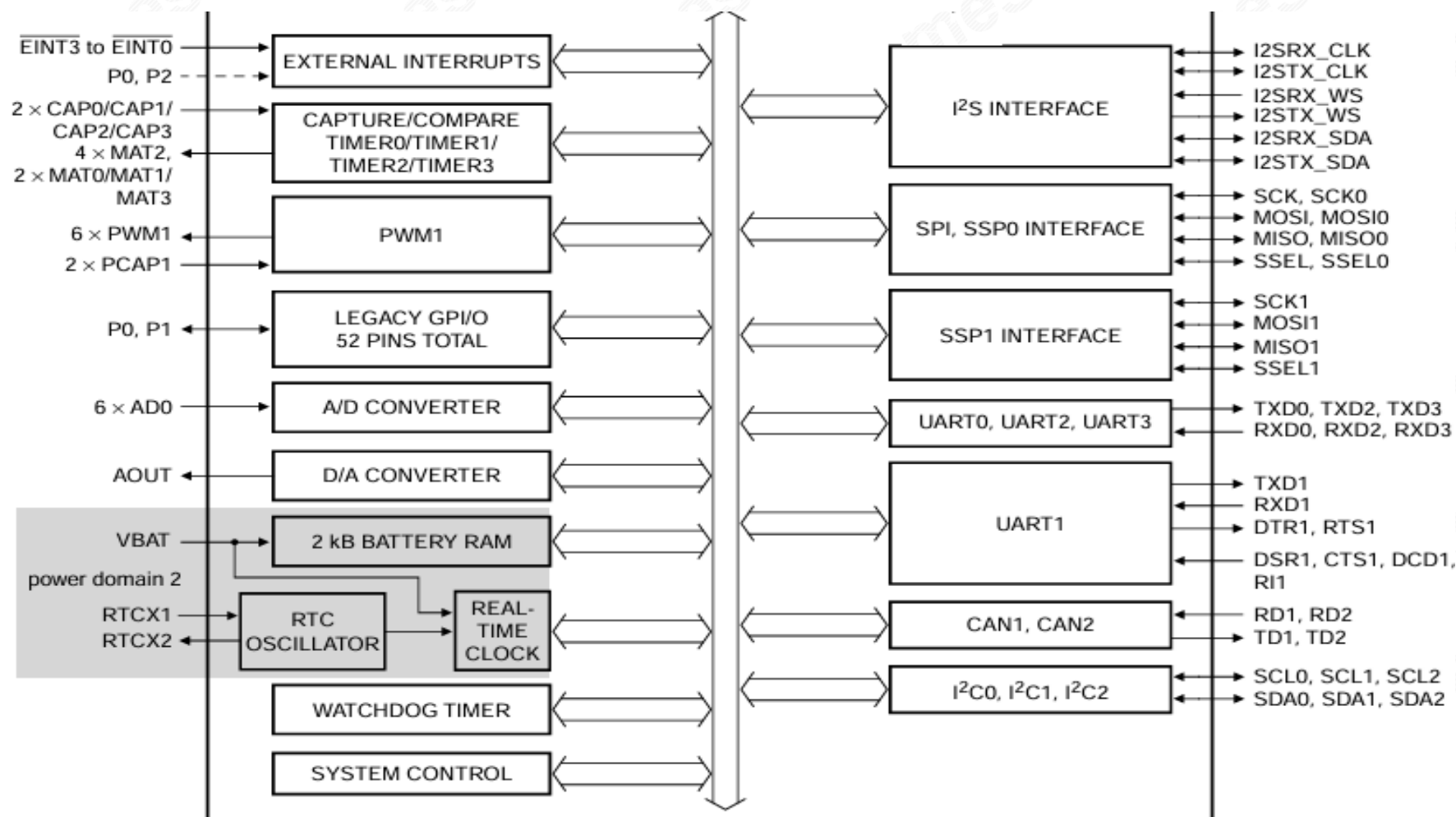


ARM Architecture

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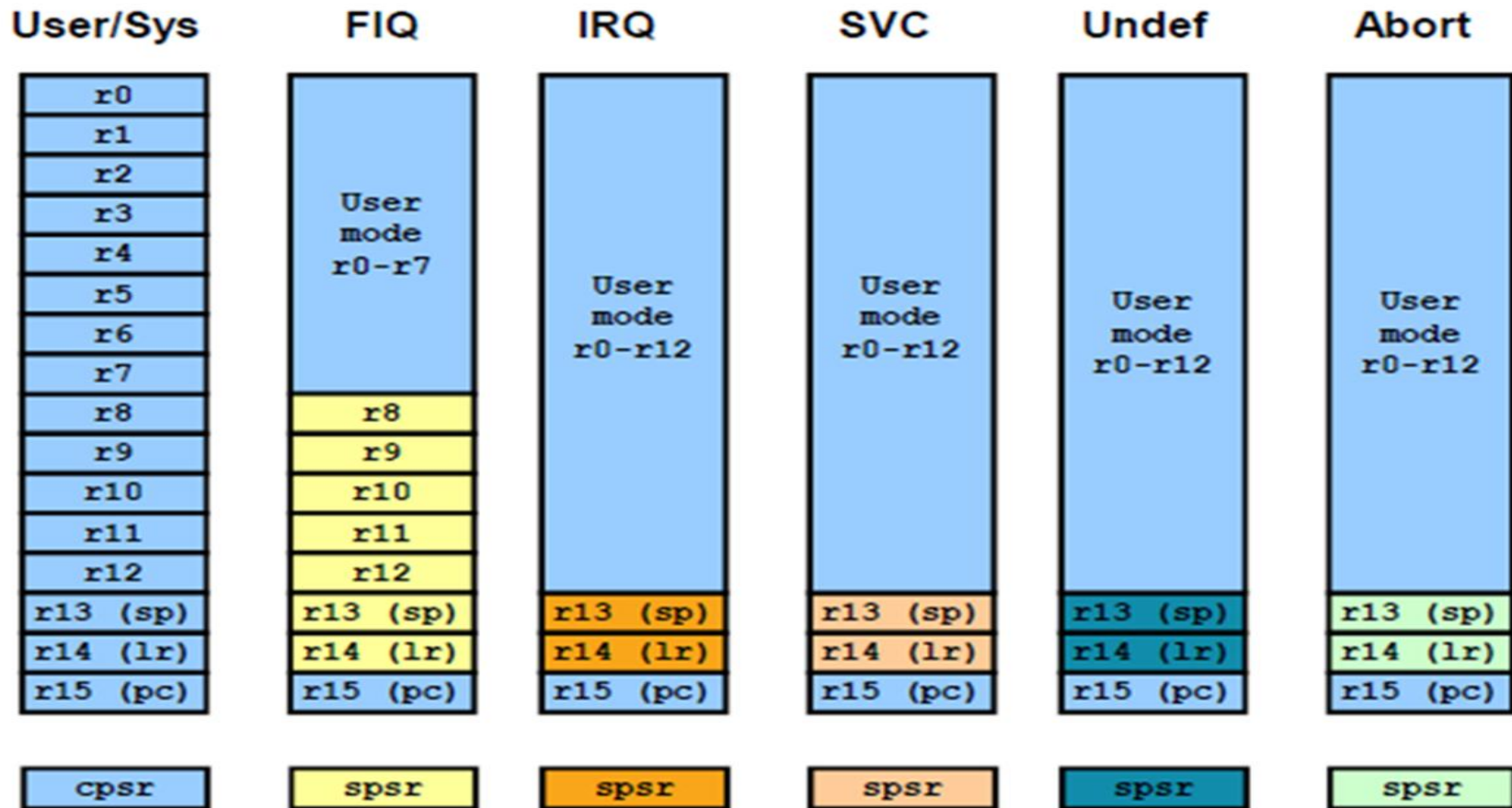
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ARM Register Set

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Note: System mode uses the User mode register set

ARM Modes of Operations

Mode	Function	Privilege
User (USR)	Programs & Application	Unprivileged
FIQ	Fast interrupt exceptions	Privileged
IRQ	Interrupt exceptions	Privileged
Supervisor (SVC)	Reset or supervisor call instructions	Privileged
Abort (ABT)	Memory exception	Privileged
Undef (UND)	Undefined instruction	Privileged
System (SYS)	OS operations	Privileged

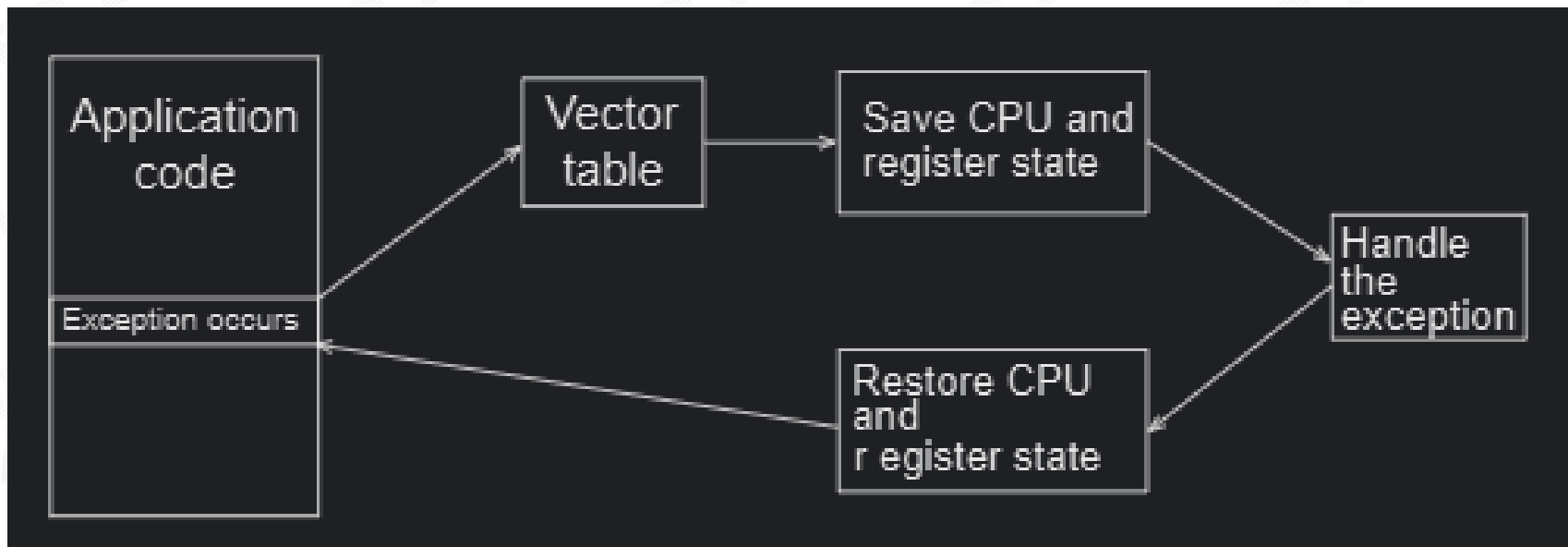
Pipelining



Fetch	Decode	Execute	Writeback
I1	-	-	-
I2	I1	-	-
I3	I2	I1	-
I4	I3	I2	I1

ARM – Exception Handling

- Try: Block in which exception occurred.
- Throw: Variable passed to catch block.
- Catch: Function to be executed on exception.



END
of
Session – 6
Thank You

